

Homework 2 – Set identities

Session 2 · Sets & set operations

HOW THIS IS MARKED

Every answer carries one sentence of justification. A correct value with no reasoning scores half. A wrong value with a clear, nearly-correct argument scores most of the marks. You may discuss problems with anyone and may use AI tools – but you must be able to explain every line you submit.

TWO FORMATS, AND THEY ARE MARKED STRICTLY

To prove an identity: double inclusion. Take an arbitrary element of one side, show it lies in the other, then do the reverse. The word "arbitrary" must appear.

To disprove one: name your sets explicitly, compute both sides, and point at the element that is in one and not the other. A Venn diagram is not a proof and is not a disproof.

Part A · Notation and computation (6 marks)

1 NOTATION

2 marks

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

2 $\in$ VS $\subseteq$

2 marks

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- $1 \in C$ (b) $\{1\} \in C$ (c) $\{1\} \subseteq C$ (d) $\{1, 2\} \subseteq C$ (e) $\emptyset \subseteq C$ (f) $|C| = 4$

3 POWER SET

2 marks

Let $A = \{a, b, c\}$.

- List $\mathcal{P}(A)$ in full, organised by the size of the subset.
- State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for any finite A .

Part B · Six identities to prove (12 marks)

FOR EVERY PROBLEM IN THIS PART

Prove by double inclusion. Each is 2 marks: 1 for each direction. A proof that verifies the identity on one example scores **zero**, however carefully the example is computed.

4 DISTRIBUTIVITY

2 marks

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

5 DE MORGAN

2 marks

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

6 DIFFERENCE

2 marks

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

7 SYMMETRIC DIFFERENCE

2 marks

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

8 SUBSETS

2 marks

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

9 CARTESIAN PRODUCT

2 marks

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .

Part C · Three to disprove, and one to write (6 marks)**10** DISPROOF

3 marks

Each statement below is **false**. For each, give explicit sets, compute both sides, and name an element that is in one side and not the other. One mark each.

1. $(A \cup B) \setminus B = A$ for all sets A, B .
2. $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ for all sets A, B .
3. $A \times B = B \times A$ for all sets A, B .

11 FIND THE ERROR

2 marks

A student submits the following. Identify the first step that is wrong, say why it is wrong, and state whether the claimed identity is nevertheless true.

Claim. $A \setminus (B \setminus C) = (A \setminus B) \cup C$ for all sets A, B, C .

"Proof." Let $x \in A \setminus (B \setminus C)$. Then $x \in A$ and $x \notin B \setminus C$. Now $x \notin B \setminus C$ means $x \notin B$ or $x \in C$. So $x \in A \setminus B$ or $x \in C$, hence $x \in (A \setminus B) \cup C$. Therefore the two sets are equal. ■

In no more than five sentences: a Venn diagram convinced you that some identity was true, and then you were asked to prove it by double inclusion anyway. What does the proof give you that the picture does not? You are not being asked to prove anything here.